



INCLUDES

- ✓ Course framework
- ✓ Instructional section
- ✓ Sample exam questions

AP[®] Physics 2: Algebra-Based

COURSE AND EXAM DESCRIPTION

Effective
Fall 2024

ADVANCED PLACEMENT PHYSICS 2 TABLE OF INFORMATION

CONSTANTS AND CONVERSION FACTORS		UNIT SYMBOLS
Avogadro's number,	$N_0 = 6.02 \times 10^{23} \text{ mol}^{-1}$	ampere, A
Universal gas constant,	$R = 8.31 \text{ J}/(\text{mol} \cdot \text{K})$	coulomb, C
Boltzmann's constant,	$k_B = 1.38 \times 10^{-23} \text{ J/K}$	degree Celsius, °C
1 atmosphere of pressure,	$1 \text{ atm} = 1.0 \times 10^5 \text{ N/m}^2 = 1.0 \times 10^5 \text{ Pa}$	electron volt, eV
Coulomb constant,	$k = \frac{1}{4\pi\epsilon_0} = 9.0 \times 10^9 \frac{\text{N} \cdot \text{m}^2}{\text{C}^2}$	farad, F
Proton mass,	$m_p = 1.67 \times 10^{-27} \text{ kg}$	hertz, Hz
Neutron mass,	$m_n = 1.67 \times 10^{-27} \text{ kg}$	joule, J
Electron mass,	$m_e = 9.11 \times 10^{-31} \text{ kg}$	kelvin, K
Elementary charge,	$e = 1.60 \times 10^{-19} \text{ C}$	kilogram, kg
Vacuum permittivity,	$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/(\text{N} \cdot \text{m}^2)$	meter, m
Vacuum permeability,	$\mu_0 = 4\pi \times 10^{-7} (\text{T} \cdot \text{m})/\text{A}$	mole, mol
1 electron volt,	$1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$	newton, N
Planck's constant,	$h = 6.63 \times 10^{-34} \text{ J} \cdot \text{s} = 4.14 \times 10^{-15} \text{ eV} \cdot \text{s}$ $hc = 1.99 \times 10^{-25} \text{ J} \cdot \text{m} = 1240 \text{ eV} \cdot \text{nm}$	ohm, Ω
Speed of light,	$c = 3.00 \times 10^8 \text{ m/s}$	pascal, Pa
Wien's constant,	$b = 2.90 \times 10^{-3} \text{ m} \cdot \text{K}$	second, s
Stefan-Boltzmann constant,	$\sigma = 5.67 \times 10^{-8} \text{ W}/(\text{m}^2 \cdot \text{K}^4)$	tesla, T
1 unified atomic mass unit,	$1 \text{ u} = 1.66 \times 10^{-27} \text{ kg} = 931 \text{ MeV}/c^2$	volt, V
Universal gravitational constant, $G = 6.67 \times 10^{-11} \text{ m}^3/(\text{kg} \cdot \text{s}^2) = 6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2$		watt, W
Magnitude of the acceleration due to gravity at Earth's surface, $g = 9.8 \text{ m/s}^2$		
Magnitude of the gravitational field strength at Earth's surface, $g = 9.8 \text{ N/kg}$		

PREFIXES		
Factor	Prefix	Symbol
10^{12}	tera	T
10^9	giga	G
10^6	mega	M
10^3	kilo	k
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p

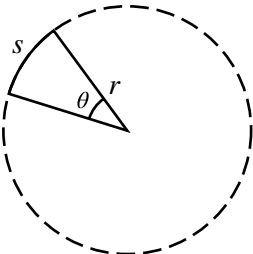
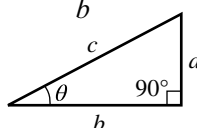
VALUES OF TRIGONOMETRIC FUNCTIONS FOR COMMON ANGLES							
θ	0°	30°	37°	45°	53°	60°	90°
$\sin \theta$	0	$1/2$	$3/5$	$\sqrt{2}/2$	$4/5$	$\sqrt{3}/2$	1
$\cos \theta$	1	$\sqrt{3}/2$	$4/5$	$\sqrt{2}/2$	$3/5$	$1/2$	0
$\tan \theta$	0	$\sqrt{3}/3$	$3/4$	1	$4/3$	$\sqrt{3}$	∞

The following conventions are used in this exam:

- The frame of reference of any problem is assumed to be inertial unless otherwise stated.
- Air resistance is assumed to be negligible unless otherwise stated.
- Springs and strings are assumed to be ideal unless otherwise stated.
- The electric potential is zero at an infinite distance from an isolated point charge.
- The direction of current is the direction in which positive charges would drift.
- All batteries, wires, and meters are assumed to be ideal unless otherwise stated.

ELECTRICITY	MAGNETISM
$ \vec{F}_E = \frac{1}{4\pi\epsilon_0} \frac{ q_1 q_2 }{r^2} = k \frac{ q_1 q_2 }{r^2}$ $\vec{E} = \frac{\vec{F}_E}{q}$ $ \vec{E} = \frac{1}{4\pi\epsilon_0} \frac{ q }{r^2} = k \frac{ q }{r^2}$ $U_E = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r} = k \frac{q_1 q_2}{r}$ $\Delta V = \frac{\Delta U_E}{q}$ $V = \frac{1}{4\pi\epsilon_0} \sum_i \frac{q_i}{r_i}$ $ \vec{E} = \left \frac{\Delta V}{\Delta r} \right $ $C = \frac{Q}{\Delta V}$ $C = \kappa\epsilon_0 \frac{A}{d}$ $E_C = \frac{Q}{\kappa\epsilon_0 A}$ $U_C = \frac{1}{2} Q \Delta V$ $I = \frac{\Delta Q}{\Delta t}$ $R = \frac{\rho \ell}{A}$ $P = I \Delta V$ $I = \frac{\Delta V}{R}$ $R_{\text{eq},s} = \sum_i R_i$ $\frac{1}{R_{\text{eq},p}} = \sum_i \frac{1}{R_i}$ $\frac{1}{C_{\text{eq},s}} = \sum_i \frac{1}{C_i}$ $C_{\text{eq},p} = \sum_i C_i$ $\tau = R_{\text{eq}} C_{\text{eq}}$	$F_B = qvB \sin \theta$ $B = \frac{\mu_0 I}{2\pi r}$ $F_B = I \ell B \sin \theta$ $\Phi_B = \vec{B} \cdot \vec{A}$ $\Phi_B = \vec{B} \cos \theta \vec{A} $ $ \mathcal{E} = \left \frac{\Delta \Phi_B}{\Delta t} \right $ $\mathcal{E} = B \ell v$ <i>A</i> = area <i>B</i> = magnetic field <i>F</i> = force <i>I</i> = current ℓ = length <i>q</i> = charge <i>r</i> = distance, radius, or position <i>t</i> = time <i>v</i> = velocity or speed $\mathcal{E}$ = emf θ = angle Φ = flux
THERMAL PHYSICS	
	$P = \frac{F_{\perp}}{A}$ $K_{\text{avg}} = \frac{3}{2} k_B T = \frac{1}{2} m v_{\text{rms}}^2$ $\frac{Q}{\Delta t} = \frac{k A \Delta T}{L}$ $PV = nRT = N k_B T$ $U = \frac{3}{2} nRT = \frac{3}{2} N k_B T$ $W = -P \Delta V$ $\Delta U = Q + W$ $Q = mc \Delta T$ <i>A</i> = area <i>c</i> = specific heat <i>F</i> = force <i>k</i> = thermal conductivity <i>K</i> = kinetic energy <i>L</i> = length <i>m</i> = mass <i>n</i> = number of moles <i>N</i> = number of atoms <i>P</i> = pressure <i>Q</i> = energy transferred to a system by heating <i>t</i> = time <i>T</i> = temperature <i>U</i> = internal energy <i>v</i> = velocity or speed <i>V</i> = volume <i>W</i> = work done on a system

WAVES, SOUND, AND OPTICS		MODERN PHYSICS	
$\lambda = \frac{v}{f}$	$a = \text{width}$	$E = hf$	$A = \text{area}$
$n = \frac{c}{v}$	$A = \text{amplitude}$	$\lambda = \frac{h}{p}$	$b = \text{constant}$
$n_1 \sin \theta_1 = n_2 \sin \theta_2$	$d = \text{separation}$	$\lambda = \frac{c}{f}$	$E = \text{energy}$
$\frac{1}{s_i} + \frac{1}{s_o} = \frac{1}{f}$	$D = \text{path length}$	$\lambda_{\max} = \frac{b}{T}$	$f = \text{frequency}$
$ M = \left \frac{h_i}{h_o} \right = \left \frac{s_i}{s_o} \right $	$f = \text{frequency or focal length}$	$P = A\sigma T^4$	$K = \text{kinetic energy}$
$\Delta D = m\lambda$	$F = \text{force}$	$K_{\max} = hf - \phi$	$m = \text{mass}$
$\Delta D = a \sin \theta$	$h = \text{height}$	$\Delta \lambda = \frac{h}{m_e c} (1 - \cos \theta)$	$N = \text{number of particles}$
$a \left(\frac{y_{\min}}{L} \right) \approx m\lambda$	$\ell = \text{length}$	$E = mc^2$	$p = \text{momentum}$
$\Delta D = d \sin \theta$	$L = \text{distance}$	$N = N_0 e^{-\lambda t}$	$P = \text{power}$
$d \left(\frac{y_{\max}}{L} \right) \approx m\lambda$	$m = \text{order or mass}$	$\lambda = \frac{\ln 2}{t_{1/2}}$	$t = \text{time}$
$v_{\text{string}} = \sqrt{\frac{F_T}{m/\ell}}$	$M = \text{magnification}$		$T = \text{absolute temperature}$
$T = \frac{1}{f}$	$n = \text{index of refraction}$		$\theta = \text{angle}$
$x(t) = A \cos(\omega t) = A \cos(2\pi f t)$	$s = \text{position}$		$\lambda = \text{wavelength or decay constant}$
$y(x) = A \cos \left(2\pi \frac{x}{\lambda} \right)$	$t = \text{time}$		$\sigma = \text{constant}$
$ f_{\text{beat}} = f_1 - f_2 $	$T = \text{period}$		$\phi = \text{work function}$
	$v = \text{speed}$		
	$x = \text{position}$		
	$y = \text{position}$		
	$\lambda = \text{wavelength}$		
	$\theta = \text{angle}$		
	$\omega = \text{angular frequency}$		

GEOMETRY AND TRIGONOMETRY			
Rectangle $A = bh$	Rectangular Solid $V = \ell wh$	$A = \text{area}$ $b = \text{base}$ $C = \text{circumference}$ $h = \text{height}$ $\ell = \text{length}$ $r = \text{radius}$ $s = \text{arc length}$ $S = \text{surface area}$ $V = \text{volume}$ $w = \text{width}$ $\theta = \text{angle}$	Right Triangle $a^2 + b^2 = c^2$ $\sin \theta = \frac{a}{c}$ $\cos \theta = \frac{b}{c}$ $\tan \theta = \frac{a}{b}$
Triangle $A = \frac{1}{2}bh$	Cylinder $V = \pi r^2 \ell$ $S = 2\pi r \ell + 2\pi r^2$		
Circle $A = \pi r^2$ $C = 2\pi r$ $s = r\theta$	Sphere $V = \frac{4}{3}\pi r^3$ $S = 4\pi r^2$		

MECHANICS AND FLUIDS

$$v_x = v_{x0} + a_x t$$

$$x = x_0 + v_{x0} t + \frac{1}{2} a_x t^2$$

$$v_x^2 = v_{x0}^2 + 2a_x(x - x_0)$$

$$\vec{x}_{\text{cm}} = \frac{\sum m_i \vec{x}_i}{\sum m_i}$$

$$\vec{a}_{\text{sys}} = \frac{\sum \vec{F}}{m_{\text{sys}}} = \frac{\vec{F}_{\text{net}}}{m_{\text{sys}}}$$

$$|\vec{F}_g| = G \frac{m_1 m_2}{r^2}$$

$$|\vec{F}_f| \leq |\mu \vec{F}_n|$$

$$\vec{F}_s = -k\Delta\vec{x}$$

$$a_c = \frac{v^2}{r}$$

$$K = \frac{1}{2} m v^2$$

$$W = F_{\parallel} d = F d \cos \theta$$

$$\Delta K = \sum W_i = \sum F_{\parallel i} d_i$$

$$U_s = \frac{1}{2} k (\Delta x)^2$$

$$U_G = -\frac{G m_1 m_2}{r}$$

$$\Delta U_g = m g \Delta y$$

$$P_{\text{avg}} = \frac{W}{\Delta t} = \frac{\Delta E}{\Delta t}$$

$$P_{\text{inst}} = F_{\parallel} v = F v \cos \theta$$

$$\vec{p} = m \vec{v}$$

$$\vec{F}_{\text{net}} = \frac{\Delta \vec{p}}{\Delta t} = m \frac{\Delta \vec{v}}{\Delta t} = m \vec{a}$$

$$\vec{J} = \vec{F}_{\text{avg}} \Delta t = \Delta \vec{p}$$

$$\vec{v}_{\text{cm}} = \frac{\sum \vec{p}_i}{\sum m_i} = \frac{\sum m_i \vec{v}_i}{\sum m_i}$$

a = acceleration

d = distance

E = energy

F = force

J = impulse

k = spring constant

K = kinetic energy

m = mass

p = momentum

P = power

r = radius, distance, or position

t = time

U = potential energy

v = velocity or speed

W = work

x = position

y = height

θ = angle

μ = coefficient of friction

$$\omega = \omega_0 + \alpha t$$

$$\theta = \theta_0 + \omega_0 t + \frac{1}{2} \alpha t^2$$

$$\omega^2 = \omega_0^2 + 2\alpha(\theta - \theta_0)$$

$$v = r\omega$$

$$a_r = r\alpha$$

$$\tau = r_{\perp} F = r F \sin \theta$$

$$I = \sum m_i r_i^2$$

$$I' = I_{\text{cm}} + M d^2$$

$$\alpha_{\text{sys}} = \frac{\sum \tau}{I_{\text{sys}}} = \frac{\tau_{\text{net}}}{I_{\text{sys}}}$$

$$K = \frac{1}{2} I \omega^2$$

$$W = \tau \Delta \theta$$

$$L = I \omega$$

$$L = r m v \sin \theta$$

$$\Delta L = \tau \Delta t$$

$$\Delta x_{\text{cm}} = r \Delta \theta$$

$$T = \frac{1}{f}$$

$$T_s = 2\pi \sqrt{\frac{m}{k}}$$

$$T_p = 2\pi \sqrt{\frac{\ell}{g}}$$

$$x = A \cos(2\pi f t)$$

$$x = A \sin(2\pi f t)$$

$$\rho = \frac{m}{V}$$

$$P = \frac{F_{\perp}}{A}$$

$$P = P_0 + \rho g h$$

$$P_{\text{gauge}} = \rho g h$$

$$F_b = \rho V g$$

$$A_1 v_1 = A_2 v_2$$

$$P_1 + \rho g y_1 + \frac{1}{2} \rho v_1^2 = P_2 + \rho g y_2 + \frac{1}{2} \rho v_2^2$$

a = acceleration

A = amplitude or area

d = distance

f = frequency

F = force

h = height

I = rotational inertia

k = spring constant

K = kinetic energy

ℓ = length

L = angular momentum

m = mass

M = mass

P = pressure

r = radius, distance, or position

t = time

T = period

v = velocity or speed

V = volume

W = work

x = position

y = vertical position

α = angular acceleration

θ = angle

ρ = density

τ = torque

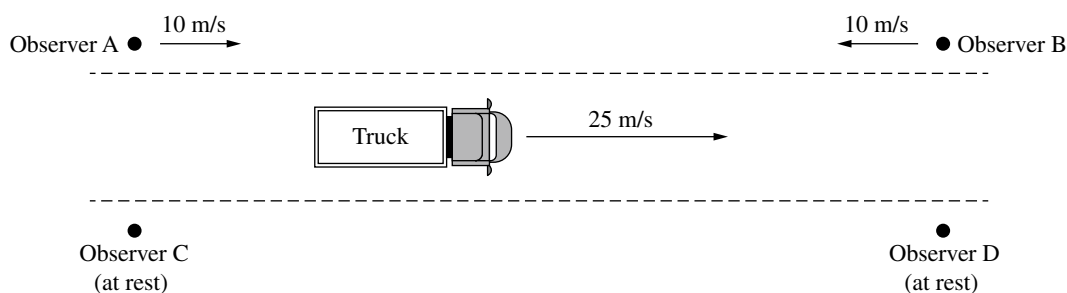
ω = angular speed

Sample Exam Questions

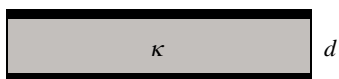
The sample exam questions that follow illustrate the relationship between the course framework and the AP Physics 2 Exam and serve as examples of the types of questions that appear on the exam. These sample questions do not represent the full range and distribution of items on an official AP Physics 2 Exam. After the sample questions is a table which shows which skill, learning objective, and essential knowledge statement each question relates to. This table also provides the answers to the multiple-choice questions.

Section I: Multiple-Choice Questions

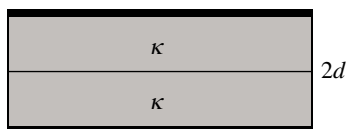
1. The root-mean-square speed of the atoms of a monatomic ideal gas at a temperature of 200 K is v_{rms} . At a later time, the root-mean-square speed of the atoms is $2v_{\text{rms}}$. What is the temperature of the gas at this later time?
(A) 800 K
(B) 400 K
(C) 200 K
(D) 50 K



2. A truck is traveling to the east with a speed of 25 m/s relative to the road while emitting a loud sound with a single frequency. The figure shows four different observers and indicates their locations and velocities relative to the road. Which observer hears a sound with the highest frequency?
(A) A
(B) B
(C) C
(D) D



Capacitor 1



Capacitor 2

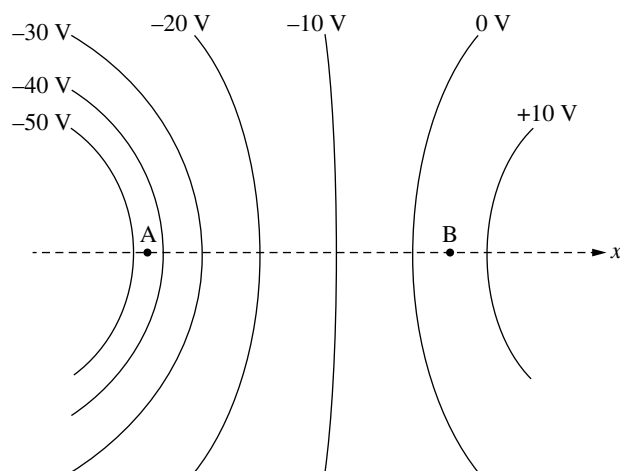
3. The figures show two different capacitors made of identical parallel conducting plates. In Capacitor 1, the plates are separated by a distance d and a block of material with dielectric constant κ is inserted between the plates. In Capacitor 2, the plates are separated by a distance $2d$ and two identical blocks of material with dielectric constant κ are inserted between the plates. A student claims that the capacitance of Capacitor 2 is the same as that of Capacitor 1. Which of the following indicates whether the claim is correct or incorrect, and provides a valid justification?

- (A) Correct, because the effects of doubling the distance between the plates and doubling the volume of the dielectric cancel each other.
- (B) Correct, because the capacitance is a property of the plates and the plates have not changed.
- (C) Incorrect, because doubling the thickness of the dielectric material will cause the capacitance to double.
- (D) Incorrect, because doubling the distance between the plates will reduce the capacitance by half.

4. At time $t = 0$, a sample of radioactive material with decay constant λ has N_0 nuclei. At time $t = t_1$, there are N_1 nuclei remaining. At a later time $t = t_2$, there are N_2 nuclei remaining. Which of the following correctly relates N_2 to N_1 ?

- (A) $N_2 = N_1 e^{-\lambda(t_2 - t_1)}$
- (B) $N_2 = N_1 e^{-\lambda(t_2 + t_1)}$
- (C) $N_2 = \frac{N_1}{N_0} e^{-\lambda(t_2 - t_1)}$
- (D) $N_2 = \frac{N_1}{N_0} e^{-\lambda(t_2 + t_1)}$

5. A student places a lens 40 cm from an object and observes an enlarged image on a screen on the opposite side of the lens. Which of the following could be the type of lens and the focal length of the lens?
- (A) A diverging lens with a focal length of 30 cm
 (B) A diverging lens with a focal length of 50 cm
 (C) A converging lens with a focal length of 30 cm
 (D) A converging lens with a focal length of 50 cm

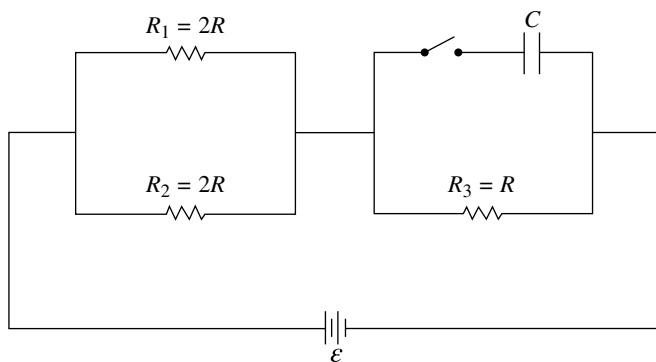


6. The figure shows equipotential lines in a particular region. The value of the potential along each line is indicated. Two points, A and B, lie along the x -axis. The electric field at Point A is $\vec{E}_A$ and the electric field at Point B is $\vec{E}_B$. Which of the following correctly compares the magnitudes of $\vec{E}_A$ and $\vec{E}_B$, and indicates whether the vectors are in the same or opposite direction?

	Magnitudes	Directions
A	$E_A < E_B$	Same direction
B	$E_A < E_B$	Opposite directions
C	$E_A > E_B$	Same direction
D	$E_A > E_B$	Opposite directions

- (A) A
 (B) B
 (C) C
 (D) D

Questions 7 and 8 refer to the following.



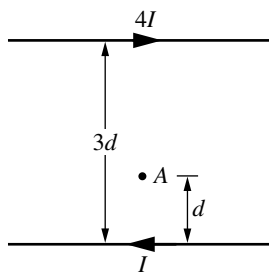
The circuit shown consists of an ideal battery of emf ε , three resistors with the resistances indicated, a capacitor of capacitance C , and a switch. The capacitor is initially uncharged and the switch is initially open.

7. When the switch is open, which of the following correctly compares the powers P_1 , P_2 , and P_3 dissipated by the resistors with resistances R_1 , R_2 , and R_3 , respectively?
- (A) $P_1 = P_2 = P_3$
- (B) $P_3 > (P_1 = P_2)$
- (C) $(P_1 = P_2) > P_3 > 0$
- (D) $(P_1 = P_2) > (P_3 = 0)$
8. The switch is closed at time $t = t_1$. A long time later, at $t = t_2$, the circuit reaches steady state conditions. Which of the following correctly indicates how the current I_s in the switch and the magnitude of the potential difference $|\Delta V_C|$ across the capacitor change between t_1 and t_2 ?

	I_s	$ \Delta V_C $
A	Increases	Increases
B	Increases	Decreases
C	Decreases	Increases
D	Decreases	Decreases

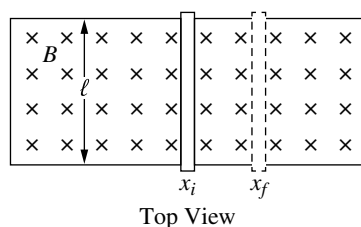
- (A) A
- (B) B
- (C) C
- (D) D

9. A 0.050 kg piece of an unknown metal is initially at a temperature of 373 K. The metal is then quickly moved into an insulated container with 0.100 kg of water at a temperature of 295 K. The water and the metal come to equilibrium at a temperature of 300 K. The specific heat of water is $4184 \text{ J}/(\text{kg} \cdot \text{K})$. What is the specific heat of the unknown metal?
- (A) $287 \text{ J}/(\text{kg} \cdot \text{K})$
 (B) $573 \text{ J}/(\text{kg} \cdot \text{K})$
 (C) $2090 \text{ J}/(\text{kg} \cdot \text{K})$
 (D) $6620 \text{ J}/(\text{kg} \cdot \text{K})$



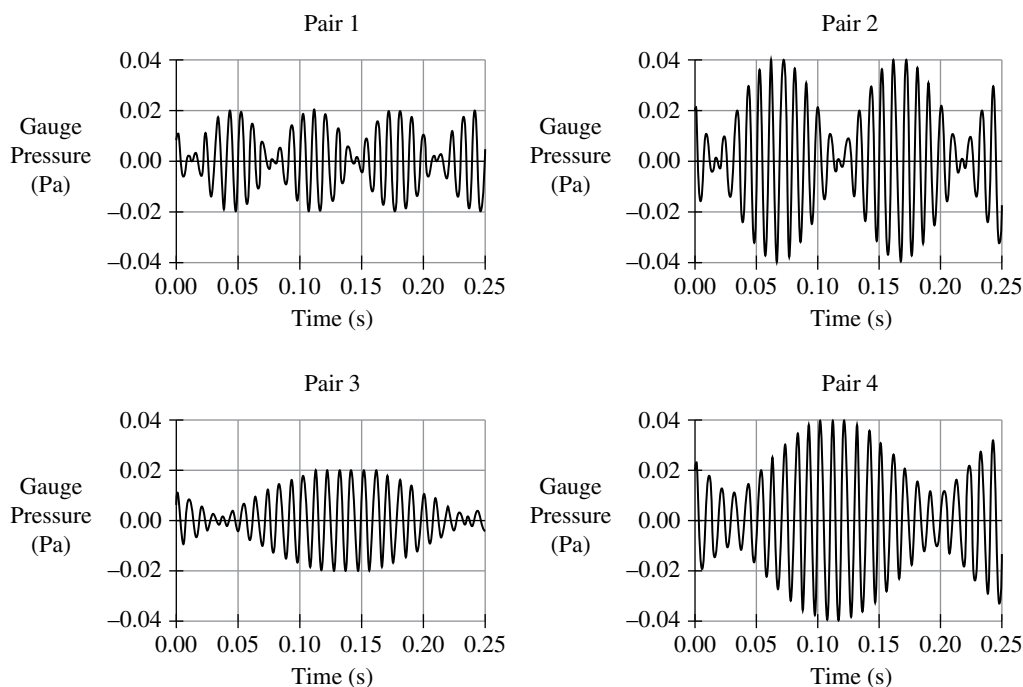
10. Two long parallel wires, each in the plane of the page, are separated by a distance $3d$, as shown. The top wire carries current $4I$ directed to the right, and the bottom wire carries current I directed to the left. Point A is also in the plane of the page, located a distance d above the bottom wire. Which of the following correctly represents the magnitude of the magnetic field due to the wires at Point A?
- (A) $\frac{1}{2} \frac{\mu_0 I}{\pi d}$
 (B) $\frac{3}{4} \frac{\mu_0 I}{\pi d}$
 (C) $\frac{3}{2} \frac{\mu_0 I}{\pi d}$
 (D) $\frac{5}{2} \frac{\mu_0 I}{\pi d}$

11. A blackbody of temperature T_0 emits a spectrum of light with a peak wavelength λ_0 . The rate at which energy is emitted by the blackbody is P_0 . The temperature of the blackbody is then changed so that the new peak wavelength emitted by the blackbody is $\frac{\lambda_0}{2}$. Which of the following correctly indicates the rate at which energy is emitted by the blackbody at this new temperature?
- (A) $\frac{P_0}{2}$
- (B) $\frac{P_0}{16}$
- (C) $2P_0$
- (D) $16P_0$



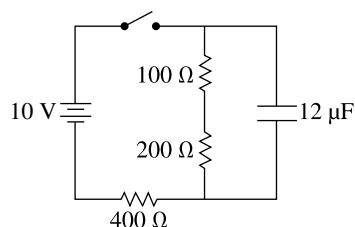
12. A metal bar of mass M and resistance R is pushed to the right at a constant speed along two horizontal, parallel, conducting rails, as shown in the top view diagram. The rails are separated by a distance ℓ , and are connected at the left end by a wire. The rails and wire have negligible resistance. The entire apparatus is in a uniform magnetic field of magnitude B that is directed into the page. At time t_i the bar is at position x_i , and a short time later at time t_f the bar is at position x_f . Which of the following expressions is equal to the average force exerted on the bar by the magnetic field between t_i and t_f ?

- (A) $\frac{\ell B}{R} \left(\frac{x_f - x_i}{t_f - t_i} \right)$
- (B) $\frac{\ell^2 B^2}{R} \left(\frac{x_f - x_i}{t_f - t_i} \right)$
- (C) $\frac{x_i B}{R} \left(\frac{x_f - x_i}{t_f - t_i} \right)$
- (D) $\frac{x_i \ell B^2}{R} \left(\frac{x_f - x_i}{t_f - t_i} \right)$

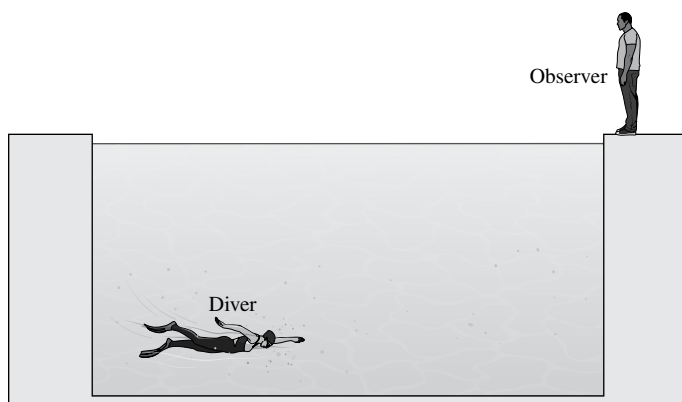


13. Pairs of tuning forks are struck near a microphone, which is connected to a computer that creates graphs of the resulting gauge pressure (the difference between the pressure and atmospheric pressure) as functions of time. The figure shows graphs for four pairs of tuning forks, where each pair consists of tuning forks that vibrate at different frequencies. Which pair of tuning forks has the greatest difference in frequencies?

- (A) Pair 1
- (B) Pair 2
- (C) Pair 3
- (D) Pair 4



14. A 10 V battery, resistors with resistances of $100\ \Omega$, $200\ \Omega$, and $400\ \Omega$, a $12\ \mu\text{F}$ capacitor, and a switch are connected as shown. The capacitor is initially uncharged and the switch is initially open. What is the current in the battery immediately after the switch is closed?
- (A) Zero
 (B) 14 mA
 (C) 25 mA
 (D) 33 mA



15. A diver is swimming underwater in a pool filled with water. An observer is standing at the edge of the pool, as shown. Will the observer always be able to see the diver no matter where the diver is in the pool? Why or why not?
- (A) Yes, because there will always be a light ray originating from the diver that will refract toward the observer and reach the observer.
 (B) Yes, because for every position of the diver, there is a straight-line path connecting the diver and the observer.
 (C) No, because light passing from a medium with high index of refraction to low index of refraction will always totally reflect at the boundary between the two media.
 (D) No, because for some positions of the diver some of the light rays from the diver will totally reflect at the water-air boundary.

Section II: Free-Response Questions

FREE-RESPONSE QUESTION: MATHEMATICAL ROUTINES

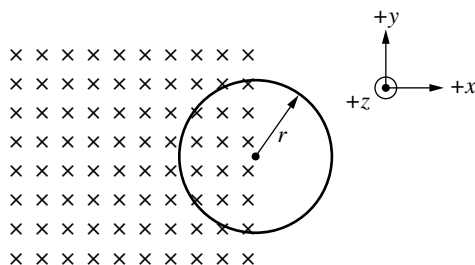


Figure 1

1. A circular conducting loop of radius r is held at rest in the xy -plane, as shown in Figure 1. The resistance of the loop is R . Half of the loop is in a region with a uniform magnetic field of magnitude B_0 that is directed into the page ($-z$ -direction). During a time interval Δt , the magnitude of the magnetic field is increased at a constant rate from B_0 to $4B_0$.
 - (a) Express your answers for (a)(i) and (a)(ii) in terms of r , R , B_0 , Δt , and physical constants, as appropriate.
 - i. **Derive** an expression for the emf $\mathcal{E}$ induced in the loop during the interval Δt . Begin your derivation by writing a fundamental physics principle or an equation from the reference information.
 - ii. **Derive** an expression for the total energy dissipated by the loop during the interval Δt . Begin your derivation by writing a fundamental physics principle or an equation from the reference information.

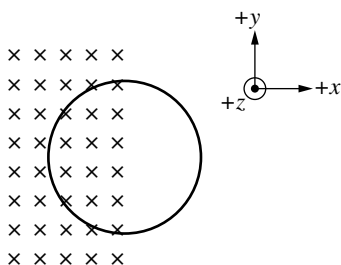


Figure 2

- iii. Complete the following tasks on Figure 2:
 - **Draw** an arrow labeled I to indicate the direction of the current in the loop during the interval Δt .
 - **Draw** an arrow labeled F to indicate the direction of the net magnetic force exerted on the loop during the interval Δt .

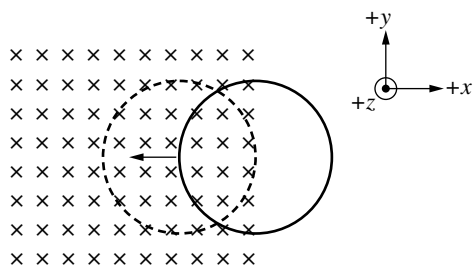


Figure 3

The magnitude of the magnetic field is now held constant while the loop is pulled in the $-x$ -direction by an external force until the loop is completely inside the region with the magnetic field, as indicated by the dashed line in Figure 3.

- (b) On both blanks below, **write** “yes” or “no” to indicate whether an emf is induced in the loop when the loop is partially in the region with the magnetic field and when the loop is completely in the region.

_____ When the loop is partially in the region

_____ When the loop is completely in the region

Justify your answers using physics principles.

FREE-RESPONSE QUESTION: TRANSLATION BETWEEN REPRESENTATIONS

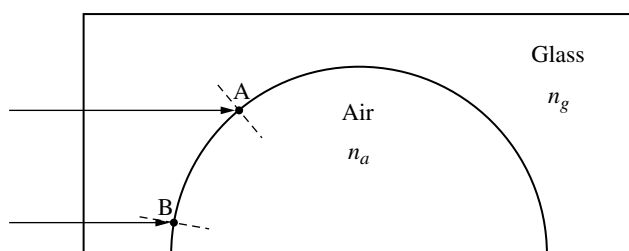


Figure 1

2. A glass block has a semicircular section removed, as shown in Figure 1. The air has an index of refraction $n_a = 1$ and the glass has an index of refraction $n_g > 1$. Two rays of monochromatic light of frequency f_0 travel parallel to each other in the glass block and are incident on the glass-air boundary at points A and B, as shown. Both rays refract at the glass-air boundary.

- (a) On the diagram of the block in Figure 2 below, **sketch** the paths of the rays in the air after the rays exit the glass at points A and B.

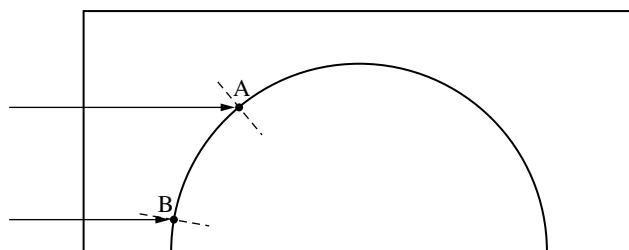


Figure 2

- (b) **Derive** an equation for the change in wavelength $\Delta\lambda$ of the light when it passes from the glass into the air. Express your answer in terms of f_0 , n_g , and fundamental constants, as appropriate. Begin your derivation by writing a fundamental physics principle or an equation from the reference information.

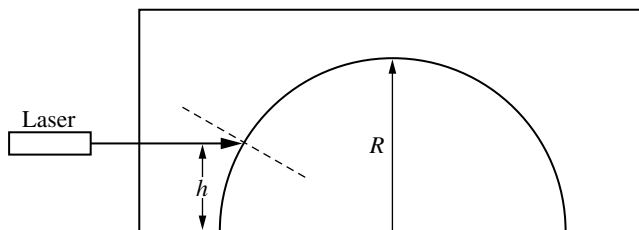


Figure 3

- (c) A laser is used to shine a horizontal beam of light on the glass block, as shown in Figure 3. The height h of the beam is varied and the angle θ_r of the refracted beam to the normal is measured for several values of h , and the value of $\sin\theta_r$ is calculated.

On the axes below in Figure 4, **sketch** $\sin\theta_r$ as a function of h .

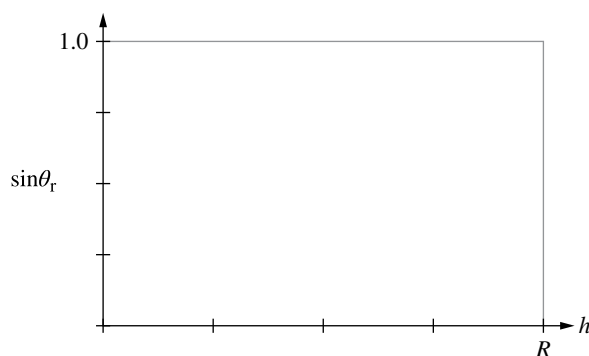
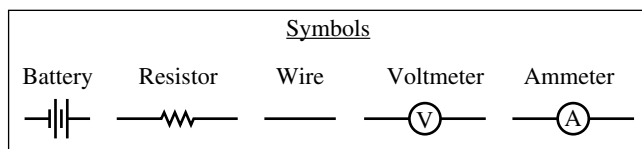


Figure 4

- (d) **Indicate** whether the sketch drawn in part (c) is consistent with the rays drawn in part (a). Briefly **justify** your response.

FREE-RESPONSE QUESTION: EXPERIMENTAL DESIGN AND ANALYSIS (LAB)

3. A student is provided with several resistors with different known resistances, an ideal battery, a voltmeter, an ammeter, and connecting wires. The student is given an additional resistive component R_x .
- (a) The student is asked to design an experimental procedure to determine whether R_x is ohmic using the equipment provided.
- i. Using the symbols shown, **draw** a diagram of a circuit that the student could use to determine whether R_x is ohmic, including the appropriate placement of the measurement devices.



- ii. Referring to your diagram, **describe** the overall procedure, providing enough detail so that students could replicate the experiment, including any steps necessary to reduce experimental uncertainty.
- (b) **Describe** how the data collected in part (a) could be graphed and how that graph would be analyzed to determine whether R_x is ohmic.

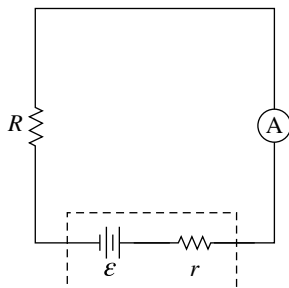


Figure 1

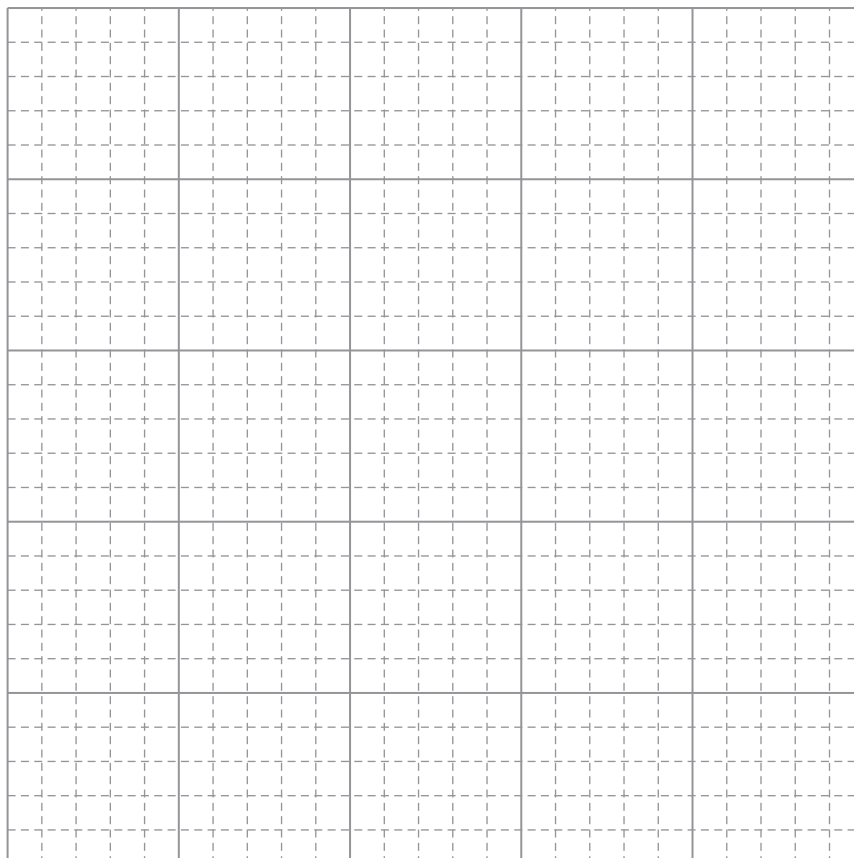
In a different experiment, the student uses the same resistors of known resistance, a battery of emf $\mathcal{E}$ with internal resistance r , and an ammeter connected as shown in Figure 1. The student is asked to determine the internal resistance r of the battery. For each value of the resistance R of the known resistors, the student measures the current in the ammeter. The student's data are represented in the following table.

$R(\Omega)$	$I(\text{mA})$			
10	58			
20	44			
30	31			
40	28			
50	22			
60	19			

- (c) i. **Indicate** two quantities that could be graphed to yield a straight line that could be used to determine the internal resistance r of the battery. Use the blank columns in the table to list any calculated quantities you will graph other than the data provided.

Vertical Axis: _____ Horizontal Axis: _____

- ii. **Plot** the data points for the quantities indicated in part (c)(i) on the graph provided. Clearly scale and label all axes, including units, as appropriate.



- iii. **Draw** a best-fit line to the data graphed in part (c)(ii).
- (d) Using the best-fit line, **calculate** the internal resistance of the battery.

**FREE-RESPONSE QUESTION: QUALITATIVE/QUANTITATIVE
TRANSLATION**

4. Monochromatic light of wavelength λ_0 is incident on a metal with work function ϕ . Electrons are ejected from the metal and have a maximum speed v_0 .
- (a) When monochromatic light with wavelength greater than λ_0 is incident on the metal, electrons are still ejected. **Indicate** whether the maximum speed of the ejected electrons will be less than, equal to, or greater than v_0 . **Justify** your response.
 - (b) **Derive** an expression for the maximum speed v_0 of the electrons when the original light of wavelength λ_0 is incident on the metal in terms of λ_0 , ϕ , and physical constants, as appropriate. Begin your derivation by writing a fundamental physics principle or an equation from the reference information.
 - (c) **Indicate** whether the equation derived in part (b) agrees with the justification in part (a). Briefly **justify** your reasoning.

Answer Key and Question Alignment to Course Framework

Multiple-Choice Question	Answer	Skill	Learning Objective	Essential Knowledge
1	A	2.D	9.1.B	9.1.B.1
2	B	3.B	14.5.A	14.5.A.2
3	D	3.C	10.6.A	10.6.A.2
4	A	2.A	15.7.B	15.7.B.2
5	C	3.B	13.4.A	13.4.A.6
6	C	2.C	10.5.B	10.5.B.2
7	B	2.C	11.4.A	11.4.A.1
8	C	3.B	11.8.B	11.8.B.2
9	B	2.B	9.5.A	9.5.A.1
10	C	2.A	12.3.A	12.3.A.1
11	D	2.D	15.4.A	15.4.A.3
12	B	2.A	12.4.A	12.4.A.3
13	A	3.B	14.6.A	14.6.A.6
14	C	2.B	11.8.B	11.8.B.2
15	A	3.C	13.3.A	13.3.A.5

Free-Response Question	Skill	Learning Objective
1	1.A, 2.A, 2.B, 3.B, 3.C	11.4.A, 12.3.B, 12.4.A
2	1.A, 1.C, 2.A, 2.D, 3.B, 3.C	13.3.A, 14.2.A, 14.3.A
3	1.B, 2.B, 2.D, 3.A	11.2.A, 11.3.B, 11.5.B
4	2.A, 2.D, 3.B, 3.C	15.1.A, 15.5.A



Scoring Guidelines

Scoring Guidelines for Question 1: Mathematical Routines

10 points

Learning Objectives: 11.4.A 12.3.B 12.4.A

- (a) i For indicating that the changing flux will induce an emf in the loop

1 point

Example response:

$$|\mathcal{E}| = \left| \frac{\Delta \Phi_B}{\Delta t} \right|$$

For correctly substituting an expression for the area of the loop into an expression for the magnetic flux through the loop

1 point

Example response:

$$\mathcal{E} = \frac{\Delta B}{\Delta t} A = \frac{\Delta B}{\Delta t} \left(\frac{1}{2} \pi r^2 \right)$$

For correctly substituting an expression for the change in the magnetic field into an expression for the magnetic flux through the loop

1 point

Example response:

$$\begin{aligned} \mathcal{E} &= \frac{(4B_0 - B_0)}{\Delta t} \left(\frac{1}{2} \pi r^2 \right) \\ &= \frac{3B_0 \pi r^2}{2\Delta t} \end{aligned}$$

- (a) ii For substituting the emf from part (a)(i) into a correct expression for power dissipated by the loop

1 point

Example response:

$$P = \frac{\mathcal{E}^2}{R} = \frac{\left(\frac{3B_0 \pi r^2}{2\Delta t} \right)^2}{R}$$

For multiplying an expression for power by the time Δt to find the total energy dissipated

1 point

Example response:

$$E = P\Delta t = \frac{9B_0^2 \pi^2 r^4}{4R\Delta t}$$

Example Response:

$$\begin{aligned}
|\mathcal{E}| &= \left| \frac{\Delta \Phi_B}{\Delta t} \right| \\
&= \frac{\Delta B}{\Delta t} A \\
&= \frac{(4B_0 - B_0)}{\Delta t} \left(\frac{1}{2} \pi r^2 \right) \\
&= \frac{3B_0 \pi r^2}{2\Delta t} \\
P &= I \Delta V = \frac{(\Delta V)^2}{R} = \frac{\mathcal{E}^2}{R} \\
P &= \frac{\left(\frac{3B_0 \pi r^2}{2\Delta t} \right)^2}{R} = \frac{9B_0^2 \pi^2 r^4}{4R\Delta t^2} \\
E &= P \Delta t = \frac{9B_0^2 \pi^2 r^4}{4R\Delta t}
\end{aligned}$$

(a) iii For drawing an arrow to indicate that the current direction is counterclockwise **1 point**

For drawing an arrow to indicate that the magnetic force is in the +x-direction (or to the right) **1 point**

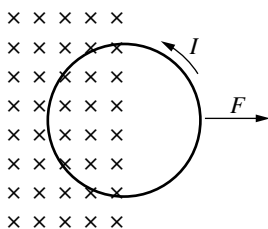
Example Response:

Figure 2

Total for part (a) 7 points

(b) For indicating “yes” for “When it is partially in the region” and “no” for “When it is completely in the region” with an attempt at a justification **1 point**

For indicating that when the loop is partially in the region, the magnetic flux is changing and therefore an emf will be induced **1 point**

For indicating that when the loop is completely in the region, the magnetic flux is constant and no emf is induced **1 point**

Example Response:

An emf will only be induced when the magnetic flux is changing. When the loop is partially in the region, the magnetic field is constant but the area of the region enclosed by the loop is changing, so the flux is changing. When the loop is completely in the region, the flux is not changing.

Total for part (b) 3 points

Total for question 1 10 points

Scoring Guidelines for Question 2: Translation Between Representations

12 points

Learning Objectives: 13.3.A 14.2.A 14.3.A

- | | | |
|-----|--|---------|
| (a) | For showing both rays refracting away from the normal or both rays refracting toward the normal | 1 point |
| | For drawing refracted rays that are refracted away from the normal relative to the incident rays | 1 point |
| | For showing the ray at Point A having a greater angle of refraction than that at Point B | 1 point |

Example Response:

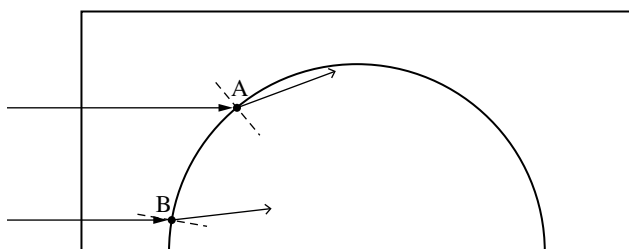


Figure 2

Total for part (a) 3 points

- | | | |
|-----|---|---------|
| (b) | For a derivation that applies the relationship between the speed, wavelength, and frequency of a wave $v = \lambda f$ | 1 point |
| | For indicating that the frequency of the light doesn't change as it passes from glass to air | 1 point |
| | For correctly expressing the wavelength within the glass using the index of refraction of the glass | 1 point |
| | For a correct equation for the change in wavelength in the light. | 1 point |

$$\Delta\lambda = \frac{c}{f_0} - \frac{c}{n_g f_0}$$

$$= \frac{c}{f_0} \left(1 - \frac{1}{n_g} \right)$$

Example Response:

$$v = \lambda f$$

$$n_g = \frac{c}{v}, \quad v = \frac{c}{n_g}$$

$$\frac{c}{n_g} = \lambda f, \quad \frac{c}{n_g f_0} = \lambda_g$$

$$c = \lambda_a f, \quad \frac{c}{f_0} = \lambda_a$$

$$\Delta\lambda = \frac{c}{f_0} - \frac{c}{n_g f_0} = \frac{c}{f_0} \left(1 - \frac{1}{n_g} \right)$$

Total for part (b) 4 points

(c)	For a monotonically increasing line or curve that starts at the origin	1 point
	For a linear function	1 point
	For a line or curve that ends at $\sin\theta_r = 1$ for $h < R$	1 point

Example Response:

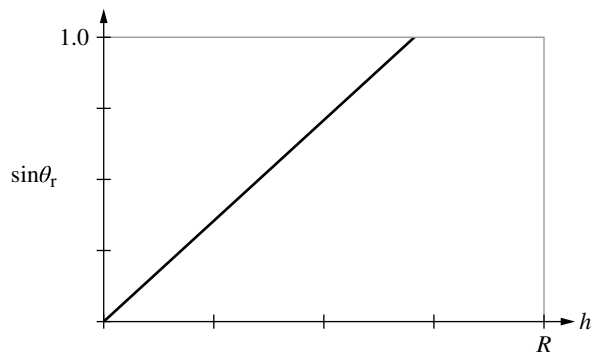


Figure 4

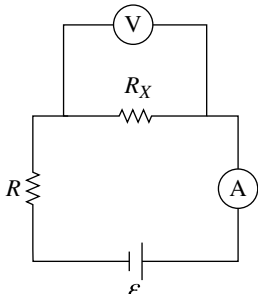
Total for part (c)		3 points
(d)	For correctly identifying whether part (c) is consistent with part (a) with an attempt at a justification using functional dependence reasoning	1 point
	For describing how one feature of the graph in part (c) is consistent with the rays drawn in part (a)	1 point
Example Response:		
<i>The graph in part (c) is consistent with the rays in part (a) because the value of h for Point A is larger than for Point B and the refracted angle is greater at Point A. The graph shows that for greater values of h the refracted angle is greater.</i>		
Total for part (d)		2 points
Total for question 2		12 points

Scoring Guidelines for Question 3: Experimental Design and Analysis 10 points

Learning Objectives: 11.2.A 11.3.B 11.5.B

- (a) i For drawing a circuit that would allow measurement of the current in and potential difference across R_X 1 point

Example Response:



- (a) ii For using multiple resistance values in a description of a procedure consistent with the diagram 1 point

Example Response:

For each value of R , use the voltmeter to measure the potential difference across R_X , and use the ammeter to measure the current in R_X . Repeat the measurements for multiple values of R .

Total for part (a) 2 points

- (b) For indicating an appropriate relationship between potential difference and current, e.g., current as a function of potential difference 1 point

For indicating that for an ohmic resistor, the relationship between potential difference and current will be linear 1 point

Example Response:

Graph current in R_X as a function of potential difference across R_X . The graph will be linear if resistor R_X is ohmic.

Total for part (b) 2 points

- (c) i For indicating quantities that would produce a straight-line graph that could be used to determine the internal resistance of the battery, e.g., resistance and the inverse of current 1 point

- (c) ii For labeling axes correctly (including units) with a linear scale such that the data fills at least half the area of the graph 1 point

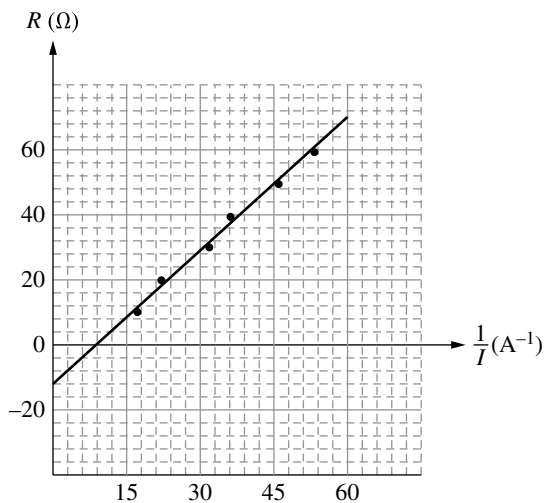
For plotting the data points correctly 1 point

- (c) iii For drawing an appropriate line or curve of best fit that approximates the data 1 point

Example Response:

Graph the resistance R of the resistors as a function of $\frac{1}{I}$.

$R \text{ } (\Omega)$	$I \text{ (mA)}$	$\frac{1}{I} \text{ (A}^{-1}\text{)}$		
10	58	17		
20	44	23		
30	31	32		
40	28	36		
50	22	46		
60	19	53		

**Total for part (c)****4 points****(d)**

For using an appropriate feature of the best-fit line to calculate the internal resistance of the battery

1 point

For calculating the internal resistance within an acceptable range

1 point**Example Response:**

$$\mathcal{E} = (r + R)I = rI + RI$$

$$R = \frac{\mathcal{E}}{I} - r$$

The y -intercept is $-r$. Therefore, $r = 12 \text{ } \Omega$.

Total for part (d)**2 points****Total for question 3****10 points**

Scoring Guidelines for Question 4: Qualitative/Quantitative Translation

8 points

Learning Objectives: 15.1.A 15.5.A

(a) For indicating that the maximum speed of the electrons will be less than v_0 , with an attempt at a justification 1 point

For indicating that the energy (or frequency) of the light decreases when the wavelength increases 1 point

For indicating that the decrease in photon energy (or frequency) leads to a decrease in maximum electron kinetic energy, and therefore a decrease in maximum speed 1 point

Example Response:

The maximum speed of the electrons will be less than v_0 . A greater wavelength means less photon energy, therefore the maximum speed of the ejected electrons will be less because the maximum kinetic energy will be less.

Total for part (a) 3 points

(b) For using $K_{\max} = hf - \phi$ with an attempt to substitute for wavelength 1 point

For a correct substitution of $f = \frac{c}{\lambda}$ 1 point

For a correct expression for $v_{\max}$: 1 point

$$v_{\max} = \sqrt{2 \left(\frac{hc}{\lambda_0} - \phi \right) \frac{1}{m_e}}$$

Example Response:

$$K_{\max} = hf - \phi$$

$$\frac{1}{2} m_e v_{\max}^2 = h \left(\frac{c}{\lambda_0} \right) - \phi$$

$$v_{\max} = \sqrt{2 \left(\frac{hc}{\lambda_0} - \phi \right) \frac{1}{m_e}}$$

Total for part (b) 3 points

(c) For a statement that addresses the functional dependence of the maximum speed on the wavelength 1 point

For a correct justification that links the functional dependence in the equation in part (b) to qualitative reasoning in part (a) 1 point

Example Response:

In part (b), my derivation shows that as wavelength increases, the value inside the square root decreases, which means v_{max} decreases. This is consistent with the claim in part (a) that light with a greater wavelength results in a smaller maximum speed of the ejected electrons.

Total for part (c)	2 points
--------------------	----------

Total for question 4	8 points
----------------------	----------
